INFORMATION TECHNOLOGY UNIVERSITY (ITU), LAHORE

Department of Data Science

ASSIGNMENT 1

Course Title: Generative AI & LLMs
Semester: Fall 2026

Assignment Topic: Text Classification (Task 3)

Submitted By

Name: Fazeelat Maheen
Program: MS Data Science (MSDS)
Roll Number: MSDS-25-040

Submitted To

Dr. Mohsin Ali

Teaching Assistant

Miss Hajra Saviour

Submission Date: October 19, 2026

PROJECT LINKS

1. Google Drive Checkpoint Link:
https://drive.google.com/file/d/1Oz_d19EsUUR7olV28UOnrhy4nR-l-1hp/view?usp=sharing

2. GitHub Repository Link:
fazeelatmaheen919435-oss/assignment-1-text-classification: Assignment 1 — Text Classification, Generative AI and LLMs, Fall 2026

3. ChatGPT Conversation Link:
https://chatgpt.com/share/6ad51078-6818-83ee-82a9-c74de569e678

1. Introduction and Goal

The objective of this assignment was to perform text classification to distinguish between real published articles and AI-generated articles. Two models, EmbeddingBag and Long Short-Term Memory (LSTM), were evaluated.The experiments investigated the effects of batch size, sequence length, and optimization algorithms on model performance.

Main Goal: To compare the models using validation accuracy, test accuracy, test loss, and runtime.

2. Dataset and Experimental Setup

The dataset contained 508 articles, divided into 358 training articles, 76 validation articles, and 74 test articles. The dataset was balanced between real published articles and AI-generated articles. Only the article text was used as model input, and the labels indicated real or generated text, not factual correctness.

The text was processed using a lowercase word-and-punctuation tokenizer. The vocabulary was built using training text only, and the original data splits remained unchanged.

Two models were evaluated: an EmbeddingBag model with mean pooling and a linear layer, and an LSTM model with an embedding layer, LSTM layer, masked mean pooling, and a linear layer. Experiments used five epochs, random seed 42, and gradient clipping of 0.1.

Section 3.1: Article length histogram

A histogram was used to examine the token lengths of the 358 training articles before truncation or padding.

- Minimum length: 60 tokens
- Median length: 441 tokens
- Mean length: 483.2 tokens
- Maximum length: 2,016 tokens
- Articles exceeding 1,024 tokens: 19 (approximately 5.3%)

The mean was higher than the median, indicating a right-skewed distribution. Most articles were shorter than 1,024 tokens, while a small number were much longer.

Statistic (training split, tokens)	Value
Articles	358
Minimum length	60
Median length	441
Mean length	483.2
Maximum length	2,016
Articles longer than 1,024 tokens	19 (about 5.3%)

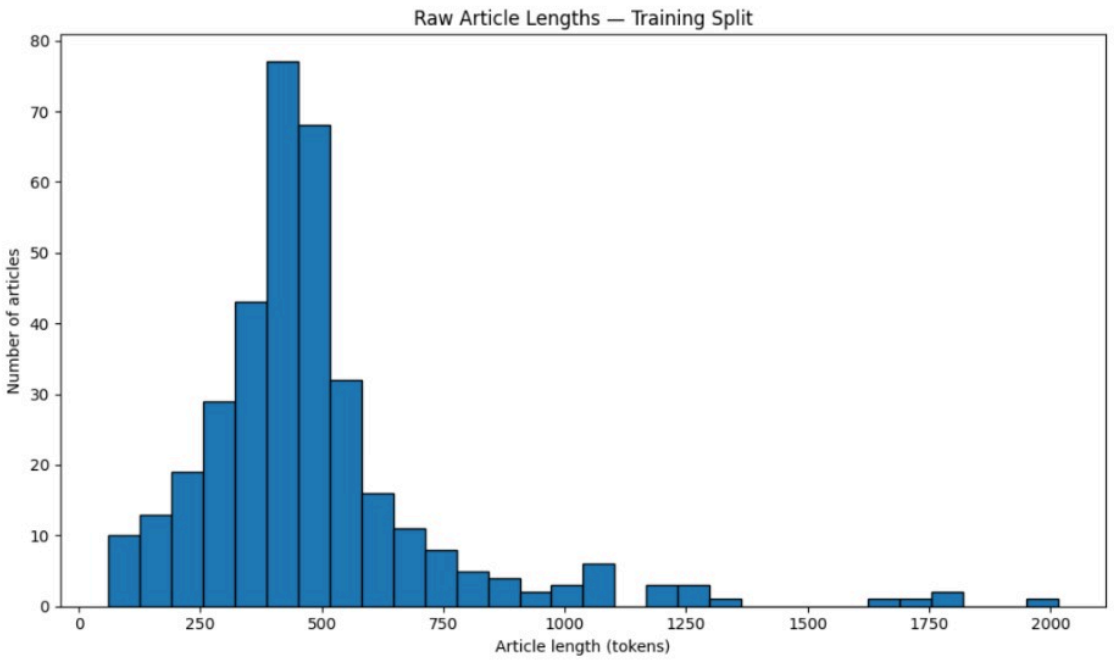


Figure 1: Raw article lengths in tokens, training split (358 articles).

Section 3.2: Truncation and padding

- Truncation limits article length, while padding makes sequences equal in length.
- Batch size 1 worked with and without truncation.
- Batch size 32 without truncation produced an error because articles had different lengths.
- Truncating to 1,024 tokens caused almost no change in test accuracy for batch size 1.
- Batch size 32 was faster but achieved lower accuracy in five epochs.

Table 1: Bag model, SGD (lr 5), 5 epochs.

Batch size	Max length	Final val acc	Test acc	Test loss	Runtime (s)
1	None (-1)	0.9474	0.9324	0.1332	7.167
1	1024	0.9342	0.9324	0.1335	7.646
32	1024	0.8553	0.8378	0.4255	2.626

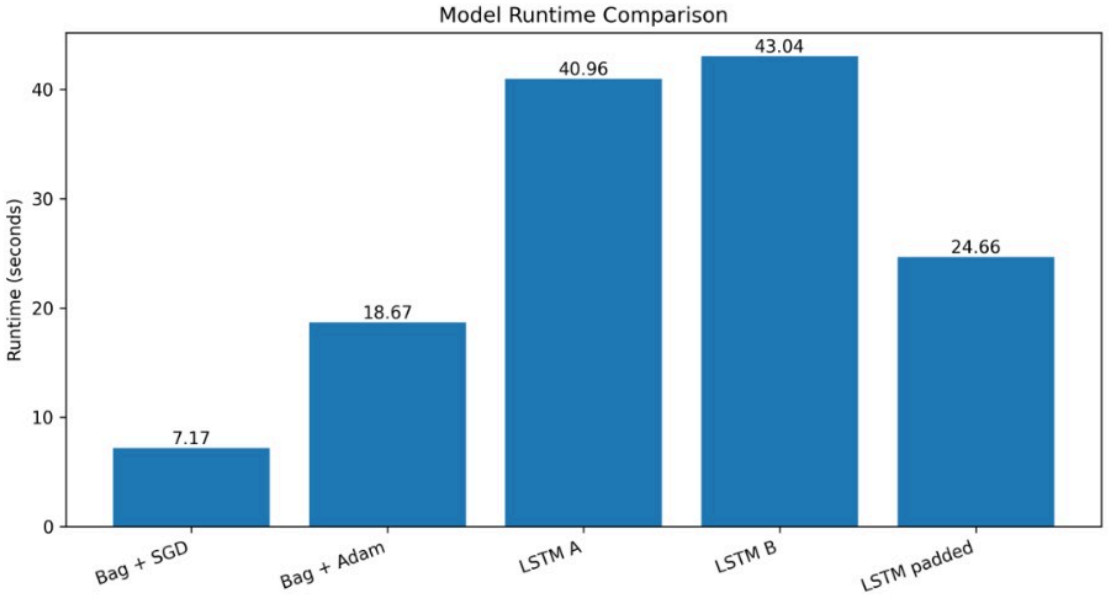


Figure 2 (optional): Runtime of the five runs in seconds.

Section 3.3: SGD vs Adam

- Adam achieved higher validation and test accuracy than SGD.
- Adam also produced a lower test loss.
- SGD completed training much faster than Adam.
- Adam performed better in these experiments, but the results are based on one run per optimizer.
- Additional runs would help confirm whether this difference is consistent.

Table 2: Bag model, batch size 1, no max length, 5 epochs.

Optimizer	Final val acc	Test acc	Test loss	Runtime (s)
SGD, lr 5	0.9474	0.9324	0.1332	7.167
Adam, default lr 0.001	0.9737	0.9865	0.0591	18.670

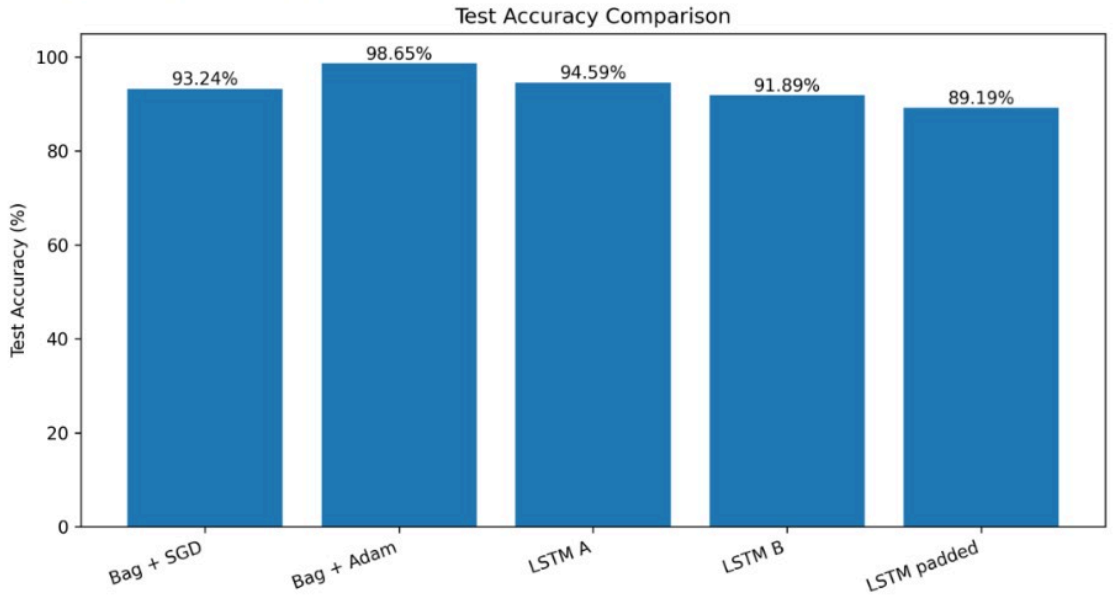


Figure 3: Test accuracy of the five runs. All runs are batch 1 with no max length except LSTM padded (batch 32, max length 1,024).

Section 3.4: LSTM

- The LSTM model used an embedding layer, an LSTM layer, masked mean pooling, and a linear layer.
- The original Bag model achieved the highest test accuracy.
- Both LSTM runs performed below the Bag model.
- High training accuracy and lower validation accuracy suggested possible overfitting.
- The padded LSTM was faster than the other LSTM runs, but its test accuracy was lower.

Table 3: Bag + Adam vs LSTM runs (Adam, default lr 0.001, 5 epochs).

Model / run	Batch	Max len	Final val acc	Test acc	Test loss	Runtime (s)
Bag + Adam	1	None	0.9737	0.9865	0.0591	18.670
LSTM run A	1	None	0.9474	0.9459	0.1872	40.959
LSTM run B (earlier)	1	None	0.9211	0.9189	0.2293	43.044
LSTM padded	32	1024	0.9079	0.8919	0.2058	24.662

LSTM run A, accuracy by epoch (train / validation):

Epoch	Train acc	Val acc
1	0.9274	0.9474
2	0.9693	0.9342
3	0.9860	0.9211
4	0.9944	0.9474
5	0.9972	0.9474

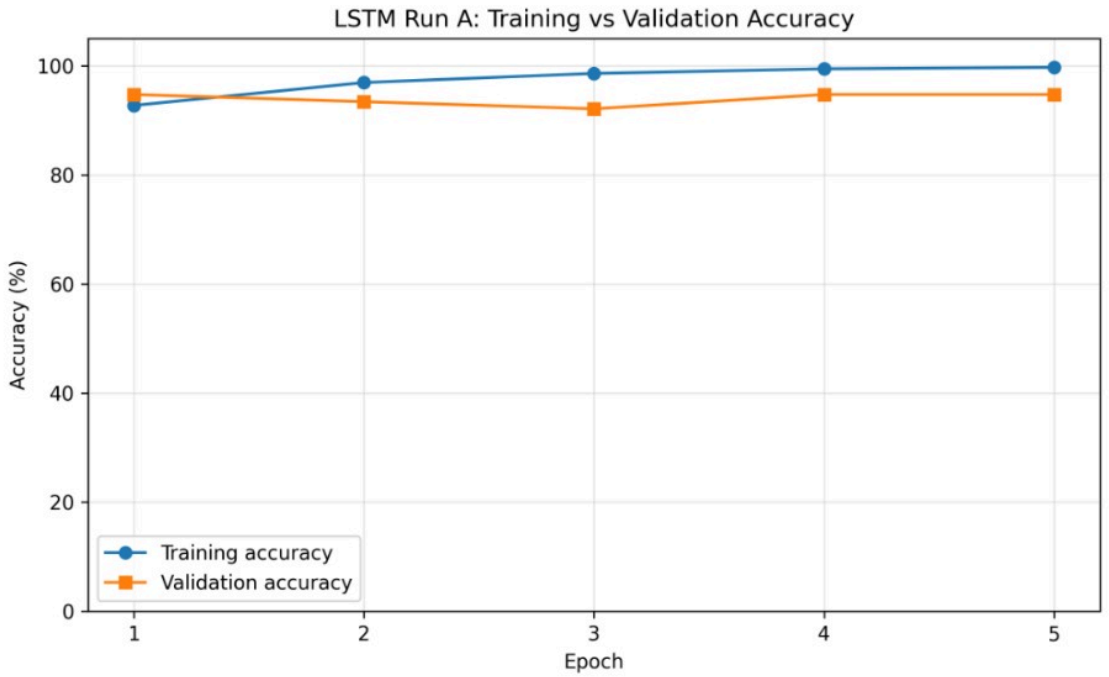


Figure 4: LSTM run A, training vs validation accuracy over 5 epochs.

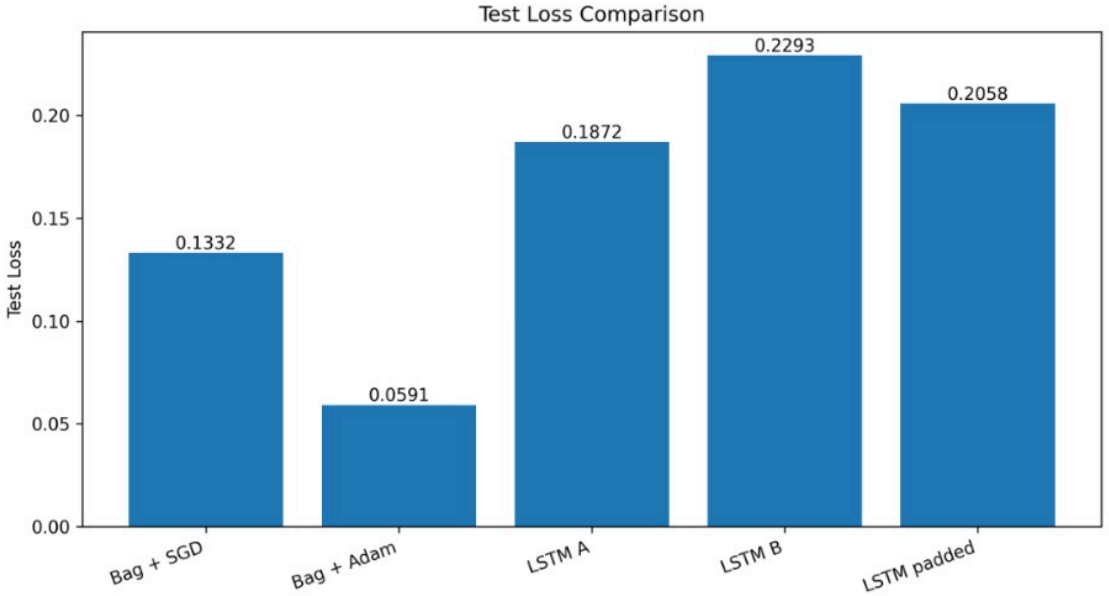


Figure 5: Test loss of the five runs (lower is better).

Section 3.5: Real vs generated text

We compared real and AI-generated articles to understand the differences in their writing styles and language. Real articles often started with specific details, such as news events, dates, places, numbers, and facts. AI-generated articles sometimes began with more general, formal, or dramatic statements.

We also noticed differences in word choice. Words such as “groundbreaking” and “unprecedented” appeared more often in the AI-generated training data than in the real articles. This suggests that AI-generated text may sometimes use broader or more expressive language, while real news articles often provide direct and specific information.

Another observation was that some AI-generated articles contained fewer specific details than real news reports. Differences in vocabulary, opening sentences, and writing style may help a classification model distinguish between real and AI-generated text.

The Bag model identifies these patterns by learning word frequencies without considering word order. The LSTM model can also learn patterns in the sequence of words. In our experiments, the Bag model performed better than the LSTM model, suggesting that word-frequency patterns were useful for this dataset.However, these findings should be interpreted carefully because only a small sample of articles was examined manually. The same patterns do not appear in every article, and no single word can reliably identify whether a text was written by a human or generated by AI. The classifier predicts whether an article is real or AI-generated based on patterns learned during training. It does not verify whether the article's information is factually correct.

Table 4: Training articles containing each word (179 articles per class).

Word	Real	Fake
groundbreaking	1	15
unprecedented	2	6
[	14	38

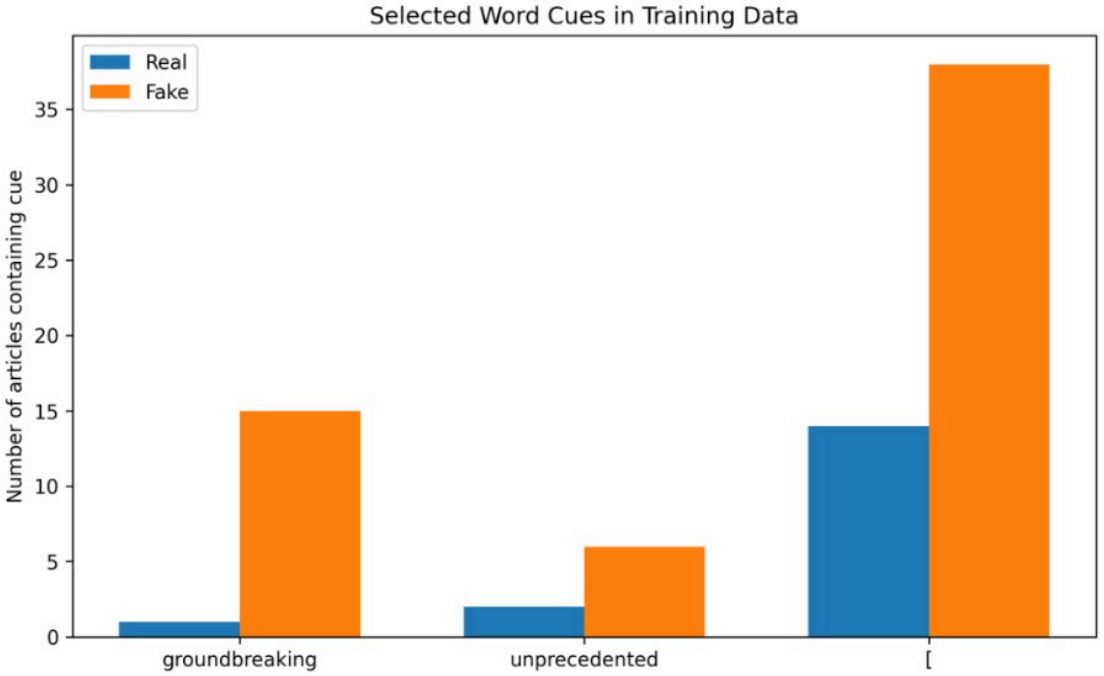


Figure 6: Number of real and fake training articles containing each word.

Limitations

- **Small dataset:** The dataset contained 508 articles, which may not represent all types of real and AI-generated text.
- **Limited validation and test data:** The validation set contained 76 articles, and the test set contained 74 articles. A small number of changed predictions could noticeably affect accuracy.
- **Limited experiments:** Most configurations were evaluated using a single run, so results may vary across repeated experiments.
- **Possible overfitting:** LSTM training accuracy reached 99.72%, while validation accuracy remained lower, suggesting possible overfitting.
- **Model performance:** The LSTM did not outperform the Bag model in our experiments.
- **Sequence length:** Truncating articles to 1,024 tokens may remove useful information from longer articles.
- **Batch-size effects:** The larger batch size was faster but achieved lower test accuracy in the five-epoch experiment.
- **Limited manual comparison:** Only a small sample of articles was examined for differences in writing style and vocabulary.
- **Dataset-specific patterns:** The classifier may learn patterns specific to this dataset and may not perform equally well on articles from other sources.
- **No factual verification:** The model classifies articles as real or AI-generated but does not verify whether their information is true.
- **Runtime variation:** Training time may differ between runs due to the execution environment and other factors.

Overall, larger datasets, repeated experiments, and testing on articles from unseen sources would help assess how well the results generalize.